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Two-Time-Scale Coordination of Vehicle-to-Grid Voltage Support and Frequency Reserve in a Two-Area System Using Proximal Policy Optimization

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ABSTRACT An aggregated fleet of vehicle-to-grid (V2G) electric vehicles can support distribution voltages and provide frequency reserve, but both services draw on the same converter rating and stored energy. This paper formulates their coordination as a two-time-scale control problem in which each area aggregator chooses, every hour, a fleet power set-point and the share of its plugged-in converter rating held as frequency reserve. The set-points are evaluated by AC power flow on an IEEE 33-bus feeder in each of two areas, together with an energy-consistent fleet model with charger ratings, efficiencies, availability, and driving demand. The reserve sets the gain and limit of a V2G droop loop in a dynamic two-area load-frequency control model with inertia, governors, a tie line, and automatic generation control, simulated at a 20-ms step. A proximal policy optimization (PPO) agent is trained with 5 independent seeds on randomized load, solar, fleet, and disturbance scenarios and compared with uncontrolled charging, a rule-based volt-watt and droop controller, and a day-ahead optimization with perfect foresight, all simulated under identical conditions. On 30 held-out days, PPO lowers the daily operating cost by 11.0%, 44.8%, and 71.6% relative to these benchmarks, reduces undervoltage by 13.5% and the mean frequency nadir by 5.0% relative to the day-ahead optimization, and leaves a smaller end-of-day energy deficit than both. The learned policy holds reserve mainly where and when vehicles are plugged in, including low-inertia midday hours, and releases it when converter capacity is needed for voltage support. The energy exchanged by the frequency droop loop is accounted for in the fleet energy balance; a comparison with exact 3600-s simulations shows that extrapolating a 60-s window to the hour overstates this energy by a factor of two to three, while the treatment adopted here tracks the exact value to within 6%, and that no treatment changes the results. Results on out-of-distribution stress days, a reward ablation, a weight-sensitivity study, and a bounded-action policy comparison are also reported, and the code, trained policies, and data are released.

INDEX TERMS Electric vehicles, frequency regulation, load-frequency control, proximal policy optimization, reinforcement learning, vehicle-to-grid, voltage regulation.

I. INTRODUCTION

Electric vehicles (EVs) connected through bidirectional chargers can return energy to the grid and adjust their charging power within seconds. This vehicle-to-grid (V2G) capability allows an aggregated fleet to take part in services that were traditionally supplied by synchronous plant and network assets [1]. Two of these services are of direct interest to network operators. At the distribution level, the active power of a fleet changes feeder flows and can therefore relieve undervoltage at heavily loaded hours or absorb surplus photovoltaic (PV) output that would otherwise raise voltages above the statutory band [10]–[12]. At the bulk-system level, fast converter control lets the same fleet respond to frequency deviations and support primary and secondary frequency control [9], [24]. The two services draw on the same converter rating and the same stored energy. A fleet that discharges at its full rating to lift an evening voltage profile has no headroom left for upward frequency response, and a fleet that has been emptied in the evening must be recharged before the next departure. The way the available capability is divided between the two services is therefore a scheduling decision in its own right.

The coupling is becoming more relevant as converter-interfaced generation displaces synchronous machines. Lower inertia raises the rate of change of frequency and deepens the frequency nadir after a disturbance, so the value of fast reserve varies over the day with the PV share [26]. The same midday hours often coincide with reverse power flow and overvoltage on PV-rich feeders, whereas the evening peak produces undervoltage on residential feeders while PV is absent. A controller that coordinates voltage support and frequency reserve must therefore act on two very different time scales: hourly energy decisions that respect network voltage limits and fleet energy, and second-level frequency dynamics that depend on how much reserve has been withheld.

The literature addresses these services largely in isolation. Studies on V2G voltage regulation solve power flow on distribution test feeders and use model predictive or robust control of charger active or reactive power [10]–[13]. Reinforcement learning (RL) has been applied to two-time-scale voltage control in active distribution systems [27], but frequency dynamics are outside the scope of those formulations. Conversely, RL methods for load-frequency control (LFC) train agents on multi-area dynamic models with tie lines and area control error (ACE) [14], [15], [17]–[19]. Several studies use V2G as the regulating resource [16], [20], [21], and an actor–critic PPO scheme has been used to mitigate EV-based load-altering attacks on microgrid LFC [22]. In these frequency-oriented formulations the fleet enters the frequency model as an aggregate power source, and the feeder voltages that result from the scheduled fleet power are not part of the evaluation. Aggregate PEV models for primary frequency control have been extended with distribution losses and current limits [24], [25], yet the allocation of fleet capability between voltage support and frequency reserve is fixed in advance rather than optimized. Planning and stability studies on EV charging infrastructure in Thai networks [3]–[7] address siting, sizing, and oscillatory stability, which are complementary to operational coordination.

This paper addresses that gap with a two-time-scale formulation in which an RL agent decides, for each area and each hour, both the V2G active-power set-point and the share of the plugged-in converter rating that is withheld as frequency-regulation reserve. The decision is evaluated on two physically specified layers. The slow layer solves an AC power flow on an IEEE 33-bus feeder [2] in each area and updates an energy-consistent fleet model. The fast layer simulates a two-area LFC system with inertia, damping, governor-turbine dynamics, a tie line with a synchronizing coefficient, ACE-based automatic generation control (AGC), and a saturated V2G droop loop whose gain and limit follow from the reserve decision. The agent is trained with proximal policy optimization (PPO) [23] and evaluated against independently simulated benchmark controllers.

The contributions are as follows.

- A two-time-scale V2G environment that couples branch-resolved AC power flow on two IEEE 33-bus feeders with a dynamic two-area LFC model in physical units. Fleet size, charger rating, battery capacity, charging and discharging efficiencies, plug-in availability, driving energy, and tie-line dynamics are specified explicitly. The power-flow solver and the LFC model are verified against pandapower [28] and against closed-form steady-state values.
- A joint action formulation in which each area aggregator selects an hourly set-point and a reserve share. The reserve share sets the droop gain and saturation of the fleet in the LFC layer, which exposes the trade-off between voltage support and frequency reserve under state-of-charge (SoC), availability, and time-varying inertia constraints. The reward penalizes both undervoltage and overvoltage, frequency nadir, tie-line deviation, battery throughput, and terminal SoC deficit.
- A reproducible learning and evaluation protocol. PPO is trained with five independent seeds on randomized load, PV, fleet participation, and disturbance scenarios. The learned policies are evaluated on 30 held-out days and 20 out-of-distribution stress days against uncontrolled charging, a conventional rule-based volt-watt and droop controller, and a day-ahead optimization with perfect foresight. A reward ablation quantifies the effect of each penalty term. Code and data are released with the paper.

The remainder of the paper is organized as follows. Section II presents the system model. Section III formulates the control problem, the PPO training procedure, and the benchmark controllers. Section IV reports the results, and Section V concludes the paper.

II. SYSTEM MODEL

A. OVERVIEW AND TIME SCALES

Fig. 1 summarizes the framework. Two interconnected control areas each contain a population of distribution feeders

hosting PV units and an aggregated V2G fleet. Each area's distribution system is represented by $N_f = 25$ identical copies of a representative IEEE 33-bus feeder, so that the area-level fleet power and reserve are N_f times the per-feeder values while the voltage response is evaluated on the representative feeder. The slow layer advances in hourly steps over a 24-h horizon. The fast layer simulates a 60-s window of frequency dynamics at a 20-ms integration step for every hour, using the reserve that the slow layer has made available. The episode starts at 12:00 and ends at 12:00 on the following day so that the evening discharge and the overnight recharge fall inside one episode.

B. DISTRIBUTION FEEDERS

Both areas use the branch impedances and nominal loads of the IEEE 33-bus system [2] at 12.66 kV, taken from the pandapower case library [28]. The five normally open tie switches are kept open so that the feeder remains radial. The net load at bus k of area i in hour t is

$$P_{k,t} = \lambda_{i,t} P_k^0 - P_{k,t}^{\mathrm{PV}} - P_{k,t}^{\mathrm{EV}}, \qquad Q_{k,t} = \lambda_{i,t} Q_k^0, \quad (1)$$

where P_k^0 and Q_k^0 are the nominal loads, $\lambda_{i,t}$ is the normalized load profile, and positive $P_{k,t}^{\mathrm{EV}}$ denotes discharge. The fleet set-point is shared equally among the V2G hubs of the area, and the PV output is shared equally among the PV buses (Fig. 2). Area 1 is a residential feeder with 1.5 MW of PV and hubs in the downstream half of the main feeder (buses 13, 17, and 18) and at the end of the bus 26 to 33 lateral. Area 2 is a mixed commercial and residential feeder with 4.2 MW of PV concentrated at the feeder ends, which produces midday overvoltage when PV output is high.

The power flow is solved by the direct backward/forward sweep of [29]. With the bus-injection to branch-current matrix $\mathbf{B}$ and the branch impedance vector $\mathbf{z}$, the bus voltages are updated as

$$\mathbf{V}^{(m+1)} = V_0 \mathbf{1} - \mathbf{B}^T \operatorname{diag}(\mathbf{z}) \mathbf{B} \left(\mathbf{S} \oslash \overline{\mathbf{V}^{(m)}} \right), \quad (2)$$

until the largest voltage update falls below 10^{-9} p.u., where $\mathbf{S}$ is the complex net load vector and $\oslash$ denotes element-wise division. The active-power loss is $\sum_b r_b |I_b|^2$ over all branches b.

C. DAILY PROFILES AND SCENARIO GENERATION

Fig. 3 shows the nominal daily profiles. The residential profile has a morning shoulder and an evening peak at 19:00 to 21:00, and the mixed profile has a broad daytime plateau and an evening peak. Both profiles are synthetic; their shapes follow the measured residential and commercial low-voltage consumer curves of [36], with a residential evening peak and a commercial daytime plateau, and are normalized to a peak of 1.0. The PV profile follows a clear-sky shape between 06:00 and 18:00. Area 1 represents home charging, with about 90% of the fleet plugged in at night and 30% during working hours, which reproduces the pattern of residential charging availability observed in the EV Project, low during the day and high overnight [37]. Area 2 represents workplace charging with the opposite pattern, following the observation that vehicles with access to workplace charging are typically parked there for 4 to 10 h on working days [38].

Each training or evaluation day is drawn from a scenario generator that scales the area load by a factor in $[0.85, 1.10]$, applies a cloud factor in $[0.3, 1.0]$ with hour-to-hour PV variability, scales fleet participation by a factor in $[0.7, 1.1]$, adds 3% multiplicative noise to the hourly load, and draws the initial SoC in $[0.50, 0.60]$. Stress days use heavier load ($[1.10, 1.20]$), lower fleet participation ($[0.55, 0.70]$), and disturbances 1.3 to 1.6 times larger, all outside the training ranges, together with near clear-sky PV ($[0.9, 1.0]$), which lies at the upper end of the training range. Stress days are never seen during training.

D. AGGREGATED EV FLEET

Each representative feeder hosts $N_{\mathrm{EV}} = 300$ vehicles with 60-kWh batteries and 7.4-kW bidirectional chargers. The usable SoC range is 0.20 to 0.90, and the charging and discharging efficiencies are $\eta_{\mathrm{ch}} = \eta_{\mathrm{dis}} = 0.92$. Each vehicle uses 9 kWh per day for driving, which corresponds to about 45 km at a real-driving consumption of about 190 Wh/km; more than 80% of European drivers cover less than 65 km per day and the median real-driving consumption of a small battery electric car was 186 Wh/km in the Green eMotion field data [39]. The driving energy is distributed over the hours in which the vehicle is away in proportion to the unplugged fraction. The fleet is treated as one aggregated battery with energy capacity $E^{\mathrm{cap}} = 18$ MWh per feeder. The plugged-in rating is

$$P_{i,t}^{\mathrm{rate}} = \chi_{i,t}\, \kappa_i\, N_{\mathrm{EV}}\, P_{\mathrm{ch}}, \quad (3)$$

where $\chi_{i,t}$ is the plugged-in fraction and κ_i the participation factor of the scenario. With the driving energy $E_{i,t}^{\mathrm{drv}}$ deducted first, the energy-feasible limits for $\Delta t = 1$ h are

$$\bar{P}_{i,t}^{\mathrm{dis}} = \min\left\{ P_{i,t}^{\mathrm{rate}},\ \eta_{\mathrm{dis}} \frac{E_{i,t} - E_{i,t}^{\mathrm{drv}} - E^{\min}}{\Delta t} \right\}^+, \quad (4)$$

$$\bar{P}_{i,t}^{\mathrm{ch}} = \min\left\{ P_{i,t}^{\mathrm{rate}},\ \frac{E^{\max} - E_{i,t} + E_{i,t}^{\mathrm{drv}}}{\eta_{\mathrm{ch}} \Delta t} \right\}^+, \quad (5)$$

and the energy is updated as

$$E_{i,t+1} = E_{i,t} - \frac{P_{i,t}^{\mathrm{dis}} \Delta t}{\eta_{\mathrm{dis}}} + \eta_{\mathrm{ch}} P_{i,t}^{\mathrm{ch}} \Delta t - E_{i,t}^{\mathrm{drv}} + \Delta E_{i,t}^{\mathrm{reg}}, \quad (6)$$

with $P^{\mathrm{dis}} = \max(P^{\mathrm{set}}, 0)$ and $P^{\mathrm{ch}} = \max(-P^{\mathrm{set}}, 0)$. If driving alone would take the energy below $E^{\min}$, the shortfall is recorded as unmet mobility energy and penalized.

The last term of (6) is the battery energy exchanged by the frequency droop loop of Section II-E. The fast layer simulates a window of length $\tau_w = 60$ s in every hour, which is too short to integrate the energy of a full hour directly. Four treatments of this term are compared in Section IV-D. Let E^{out} and E^{in} denote the energy delivered to and absorbed from the grid by the area droop power over a simulated interval, $E^{\mathrm{out}} =$

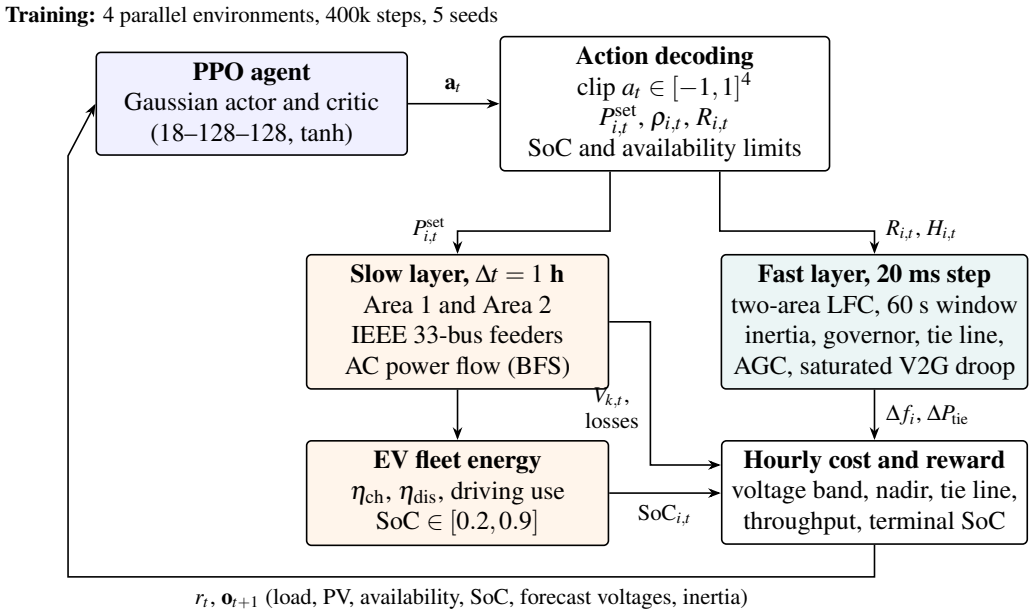


FIGURE 1. **Two-time-scale V2G coordination framework. The PPO agent sets the hourly V2G set-point and the frequency-reserve share of each area. The slow layer solves AC power flow on the IEEE 33-bus feeder of each area and updates fleet energy; the fast layer simulates two-area LFC dynamics with the reserve-dependent V2G droop.**

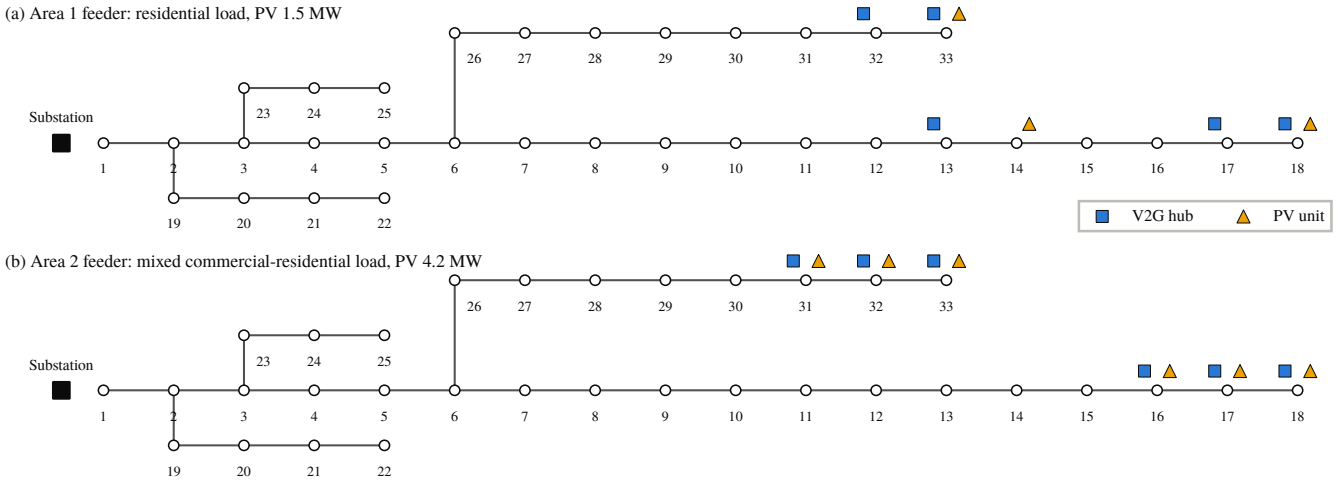


FIGURE 2. **IEEE 33-bus feeders used in the two areas, with V2G hubs and PV units. The topology and impedances are those of [2].**

$\int \max(\Delta P_{\text{ev},i}, 0)\, d\tau$ and $E^{\text{in}} = \int \max(-\Delta P_{\text{ev},i}, 0)\, d\tau$. The battery-side energy per feeder is

$$\Delta E^{\text{reg}}_{i,t} = \frac{1}{N_f}\left(-\frac{E^{\text{out}}_{i,t}}{\eta_{\text{dis}}} + \eta_{\text{ch}} E^{\text{in}}_{i,t}\right), \tag{7}$$

so that every round trip is lossy. The treatments differ in how E^{out} and E^{in} are obtained for the hour: (i) *none*, the term is neglected (the model of the first submission, used for diagnosis only); (ii) *window*, only the simulated 60 s are counted; (iii) *hourly extrapolation*, the window energy is multiplied by $3600/\tau_w = 60$, which implicitly assumes that the step disturbance of the window recurs every minute; (iv) *decomposed*, the window containing the step event is counted once and a second 60-s simulation of the same hour with the fluctuating component only is scaled to the remaining 59 min,

$$E^{\text{out}}_{i,t} = E^{\text{out}}_{\text{step+fluct}} + \left(\frac{3600}{\tau_w} - 1\right) E^{\text{out}}_{\text{fluct}}, \tag{8}$$

and likewise for E^{in}. Treatment (iv) is the default of the environment used for training and for all reported results. Its accuracy is checked against an exact 3600-s simulation of the fast layer for every hour of the evaluation (Section IV-D), which is too slow to use during training.

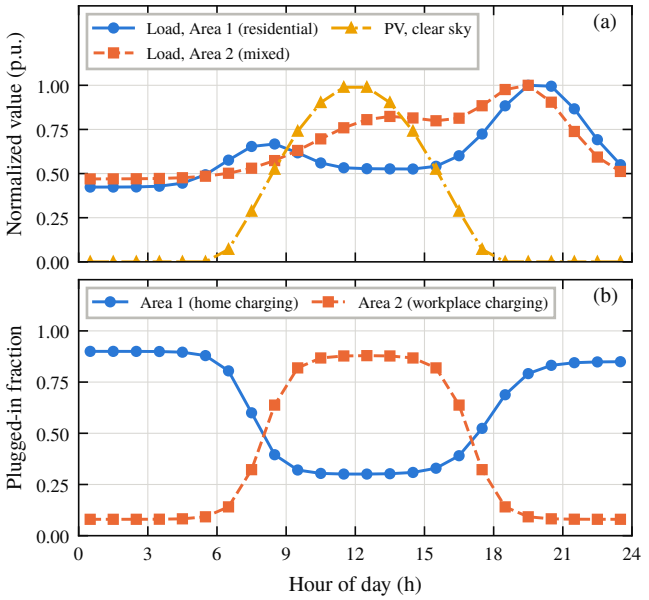


FIGURE 3. Nominal daily profiles. (a) Normalized load of each area and clear-sky PV. (b) Fraction of the fleet plugged in.

E. TWO-AREA LOAD-FREQUENCY CONTROL MODEL

The fast layer uses the standard linearized two-area LFC model [30], [31] with non-reheat turbines, extended with a V2G droop loop in each area [24]. In per unit on the area base $P_r = 500$ MW and with frequency deviation Δf_i in per unit of $f_0 = 50$ Hz,

$$2H_{i,t}\,\dot{\Delta f}_i = \Delta P_{m,i} + \Delta P_{\mathrm{ev},i} - \Delta P_{L,i} - a_i \Delta P_{\mathrm{tie}} - D_i \Delta f_i, \tag{9}$$

$$T_{g,i}\,\Delta \dot{P}_{g,i} = \Delta P_{c,i} - \Delta f_i / R_i - \Delta P_{g,i}, \tag{10}$$

$$T_{t,i}\,\Delta \dot{P}_{m,i} = \Delta P_{g,i} - \Delta P_{m,i}, \tag{11}$$

$$\Delta \dot{P}_{\mathrm{tie}} = 2\pi f_0 T_{12}\,(\Delta f_1 - \Delta f_2), \tag{12}$$

$$\mathrm{ACE}_i = B_i \Delta f_i + a_i \Delta P_{\mathrm{tie}}, \quad \Delta P_{c,i} = -K_I \int \mathrm{ACE}_i\, dt, \tag{13}$$

$$T_{\mathrm{ev}}\,\Delta \dot{P}_{\mathrm{ev},i} = \mathrm{sat}_{\pm r_{i,t}}(-K_{\mathrm{ev},i,t}\Delta f_i) - \Delta P_{\mathrm{ev},i}, \tag{14}$$

where $a_1 = 1, a_2 = -1, B_i = 1/R_i + D_i$, and $r_{i,t} = N_f R_{i,t}/P_r$ is the area V2G reserve in per unit. The droop gain is $K_{\mathrm{ev},i,t} = r_{i,t} f_0 / \Delta f_{\mathrm{full}}$, so the full reserve is deployed at $\Delta f_{\mathrm{full}} = 0.2$ Hz and the response saturates beyond that point. The first-order lag $T_{\mathrm{ev}} = 0.15$ s represents aggregator communication and converter response. Aggregated EV fleets are commonly represented in LFC studies by a first-order transfer function with time constants between 0.1 and 1 s [40], [41]; the chosen value lies at the fast end of this range; the settling time of about 1 s measured for charger response to local frequency measurements and reported in [41] corresponds to roughly three to five such time constants.

Inertia falls when PV displaces synchronous generation. The hourly inertia constant is

$$H_{i,t} = H_i^0\,(1 - \gamma_i S_{i,t})\,, \tag{15}$$

TABLE 1. Model, fleet, and training parameters.

Parameter	Value
Feeder and fleet (per representative feeder)	
Feeder / base voltage	IEEE 33-bus / 12.66 kV
PV rating, Area 1 / Area 2	1.5 / 4.2 MW
EVs, battery, charger	300, 60 kWh, 7.4 kW
SoC limits; $\eta_{\mathrm{ch}} = \eta_{\mathrm{dis}}$	0.20 to 0.90; 0.92
Driving energy per EV	9 kWh/day
Feeders per area N_f	25
Two-area LFC	
Area base P_r; f_0	500 MW; 50 Hz
H_1^0, H_2^0; γ_1, γ_2	5.0, 4.0 s; 0.20, 0.35
D_i; R_i	1.0 p.u.; 0.05 p.u.
$T_{g,1}, T_{g,2}$; $T_{t,1}, T_{t,2}$	0.08, 0.10 s; 0.40, 0.45 s
T_{12}; K_I	0.08 p.u./rad; 0.05
T_{ev}; Δf_{full}	0.15 s; 0.2 Hz
Window; integration step	60 s; 20 ms (RK4)
Load step; OU base std.	$0.03\delta u$; 0.006 p.u.
Control and reward	
Voltage band	0.95 to 1.05 p.u.
$\rho_{\max}$	0.6
$w_V, w_f, w_{\mathrm{tie}}, w_{\mathrm{deg}}, w_E, w_U$	50, 4, 0.5, 0.05, 1, 100
PPO	
Actor and critic	18–128–128, tanh
Learning rate	3×10^{-4}, linear decay to 0
Rollout; minibatch; epochs	4×384; 256; 10
Initial log σ; entropy coef.	-0.5; 0
γ; GAE λ; clip ϵ	0.995; 0.95; 0.2
Training steps; seeds	400 000; 5

where $S_{i,t}$ is the normalized PV output and $\gamma_1 = 0.20$, $\gamma_2 = 0.35$ reflect the higher PV share of Area 2 [26]. In each hour the window contains a step load change at $t = 2$ s in one randomly selected area, with magnitude $0.03\,\delta\,u$ p.u., where δ is the scenario disturbance scale and u is uniform in $[0.5, 1.0]$ with a random sign, plus an Ornstein–Uhlenbeck fluctuation with a 10-s time constant and standard deviation $0.006\,(0.3 + 2S_{i,t}(1.2 - c_i))$ p.u., where c_i is the cloud factor, so that the fluctuation grows with PV output under partial cloud. The disturbance of each hour is generated from a seed fixed by the scenario and the hour, so every controller faces identical disturbances. Equations (9) to (14) are integrated with the fourth-order Runge–Kutta method. Table 1 lists all parameters.

III. CONTROL PROBLEM AND LEARNING PROCEDURE

A. OBSERVATION AND ACTION

At the start of hour t the agent observes

$$\mathbf{o}_t = \Big[\sin\theta_t, \cos\theta_t, \lambda_{1,t}, \lambda_{2,t}, S_{1,t}, S_{2,t}, \chi_{1,t}\kappa_1, \chi_{2,t}\kappa_2, \widetilde{\mathrm{SoC}}_{1,t}, \widetilde{\mathrm{SoC}}_{2,t}, \tilde{V}^0_{\cdot,t}, \frac{H_{1,t}}{H_1^0}, \frac{H_{2,t}}{H_2^0}, \delta, \tau_t\Big], \tag{16}$$

where θ_t encodes the hour of day, $\tilde{V}^0_{\cdot,t}$ are the normalized minimum and maximum feeder voltages of both areas forecast for zero fleet power (four entries), and τ_t is the remaining fraction of the horizon, for 18 entries in total. The load, PV, and availability entries are treated as hour-ahead forecasts; the disturbance realization of the hour is not observed.

The action $\mathbf{a}_t \in \mathbb{R}^4$ sampled from the Gaussian policy is clipped to $[-1,1]^4$ before it is applied. The first two entries set the normalized set-points and the last two the reserve shares,

$$\begin{aligned} \rho_{i,t} &= \rho_{\max}\frac{a_{i+2,t}+1}{2}, \\ P^{\mathrm{set}}_{i,t} &= \Pi_{[-\bar{P}^{\mathrm{ch}}_{i,t},\,\bar{P}^{\mathrm{dis}}_{i,t}]}\left(a_{i,t}(1-\rho_{i,t})P^{\mathrm{rate}}_{i,t}\right), \end{aligned} \tag{17}$$

where Π denotes projection onto the interval. Two limiting operations are recorded separately. A raw-action clipping is counted for every component of the Gaussian action that lies outside $[-1,1]$ before clipping; because the training library clips actions before they reach the environment, this count is obtained in evaluation from the unclipped mean of the policy. A projection event is counted when the projection Π in (17) is active, that is, when a set-point within the action range is not physically feasible. The reserve that the fleet can deliver symmetrically around the set-point is

$$R_{i,t} = \min\left\{\rho_{i,t}P^{\mathrm{rate}}_{i,t},\; P^{\mathrm{rate}}_{i,t} - |P^{\mathrm{set}}_{i,t}|,\; R^{\uparrow}_{i,t},\; R^{\downarrow}_{i,t}\right\}^{+}, \tag{18}$$

where the first two terms are the requested share and the converter headroom, and $R^{\uparrow}$ and $R^{\downarrow}$ are the largest upward and downward activations that, sustained for the whole hour, keep the stored energy inside $[E^{\min}, E^{\max}]$. With the energy after the schedule $E^{\mathrm{s}}_{i,t} = E_{i,t} - E^{\mathrm{drv}}_{i,t} - P^{\mathrm{dis}}_{i,t}\Delta t/\eta_{\mathrm{dis}} + \eta_{\mathrm{ch}}P^{\mathrm{ch}}_{i,t}\Delta t$, the room above the floor $A = E^{\mathrm{s}}_{i,t} - E^{\min}$, and the room below the ceiling $B = E^{\max} - E^{\mathrm{s}}_{i,t}$,

$$R^{\uparrow}_{i,t} = \begin{cases} 0, & A \le 0, \\ A/(\eta_{\mathrm{ch}}\Delta t), & 0 < A \le \eta_{\mathrm{ch}}P^{\mathrm{ch}}_{i,t}\Delta t, \\ P^{\mathrm{ch}}_{i,t} + \eta_{\mathrm{dis}}\left(\dfrac{A}{\Delta t} - \eta_{\mathrm{ch}}P^{\mathrm{ch}}_{i,t}\right), & \text{otherwise}, \end{cases} \tag{19}$$

$$R^{\downarrow}_{i,t} = \begin{cases} 0, & B \le 0, \\ \eta_{\mathrm{dis}}B/\Delta t, & 0 < B \le \dfrac{P^{\mathrm{dis}}_{i,t}\Delta t}{\eta_{\mathrm{dis}}}, \\ P^{\mathrm{dis}}_{i,t} + \dfrac{B/\Delta t - P^{\mathrm{dis}}_{i,t}/\eta_{\mathrm{dis}}}{\eta_{\mathrm{ch}}}, & \text{otherwise}. \end{cases} \tag{20}$$

Equation (19) accounts for the two paths of an upward activation: reducing a scheduled charging power saves η_{ch} megawatt-hours of stored energy per megawatt-hour of activation, whereas activation beyond the charging power becomes a discharge that costs $1/\eta_{\mathrm{dis}}$. Equation (20) is the mirror image for a downward activation. When driving alone takes the fleet below $E^{\min}$ ($A \le 0$), no upward reserve is credited, because the charging needed to cover the mobility shortfall cannot be given up. Equations (17) to (20) guarantee that the set-point plus the droop power stays within the charger rating at every instant and that a full activation of $R_{i,t}$ in either direction, sustained for the whole hour, keeps the energy within its limits. Both properties are verified numerically in Section IV-D; the second is a design guarantee that does not depend on the length of the simulated window.

B. REWARD

The hourly reward is $r_t = -0.1\,c_t$ with

$$\begin{aligned} c_t = {} & w_V \sum_{i=1}^{2}\sum_{k=2}^{33}\left[(0.95 - V^{(i)}_{k,t})^{+} + w_{\mathrm{ov}}(V^{(i)}_{k,t} - 1.05)^{+}\right] \\ & + w_f \sum_{i=1}^{2}\frac{\max_{\tau}|\Delta f_i(\tau)|}{0.1\ \mathrm{Hz}} + w_{\mathrm{tie}}\frac{\mathrm{RMS}(\Delta P_{\mathrm{tie}})}{10\ \mathrm{MW}} \\ & + w_{\mathrm{deg}}\sum_{i=1}^{2}\frac{100|P^{\mathrm{set}}_{i,t}|\Delta t}{E^{\mathrm{cap}}} + w_U\sum_{i=1}^{2}\frac{E^{\mathrm{unmet}}_{i,t}}{E^{\mathrm{cap}}} \\ & + \mathbb{1}_{t=23}\, w_E \sum_{i=1}^{2} 100\,(\mathrm{SoC}_{i,0} - \mathrm{SoC}_{i,24})^{+}. \end{aligned} \tag{21}$$

The voltage term penalizes both undervoltage and overvoltage with $w_{\mathrm{ov}} = 1$. The frequency term uses the nadir of each area, normalized by 0.1 Hz, and the tie-line term uses the RMS deviation normalized by 10 MW. The throughput term is a linear proxy for battery wear expressed in percent of capacity per hour; energy throughput has been identified as the dominant driver of degradation for EV batteries providing V2G services [42]. The terminal term penalizes each percentage point of SoC deficit at the end of the day, so that the fleet returns to at least its initial energy. The weights were chosen so that a typical uncontrolled day produces voltage and frequency costs of the same order of magnitude; Section IV-J reports the effect of removing individual terms.

C. PPO TRAINING

The agent is trained with the PPO implementation of Stable-Baselines3 [32] using the clipped surrogate objective [23] and generalized advantage estimation [34]. Four environments run in parallel, each drawing a new random day at reset, and the hyperparameters are those of Table 1. Every 6144 environment steps, the deterministic policy (the mean of the Gaussian) is evaluated on ten fixed validation days that are disjoint from the training stream and from the test sets, and the parameters with the best validation return are retained. Training is considered converged when the validation return changes by less than 2% over the final 50 000 steps. Five independent seeds (0 to 4) are trained, and all results are reported across seeds, as recommended for RL benchmarks [35]. All policies reported in this paper, including the ablation and sensitivity variants, were trained from scratch in the final version of the environment described above; no checkpoint from an earlier model version is used.

The Gaussian policy of PPO is unbounded and the library clips sampled actions to the action box before they reach the environment. As a bounded-action alternative, a policy with a hyperbolic-tangent squashed output and generalized state-dependent exploration [33] is trained with the same budget as the ablation reference and compared in Section IV-E.

D. BENCHMARK CONTROLLERS

All benchmarks run in the same environment, with the same scenarios, disturbances, network, and constraints as PPO, and

are evaluated with the same cost (21).

Uncontrolled charging (UC): plugged-in vehicles charge at the full available rate until $\mathrm{SoC}_{\max}$ is reached, with no V2G discharge and no frequency reserve.

Rule-based control (RB): a conventional local controller. The set-point follows a volt-watt characteristic on the forecast feeder voltages, rising linearly from zero at $V_{\min} = 0.965$ p.u. to full discharge at 0.94 p.u. and falling to full charge as $V_{\max}$ rises from 1.035 to 1.05 p.u. When no voltage action is required and the SoC is below its initial value plus 0.1, the fleet charges at full rate during 00:00 to 06:00 (and during 09:00 to 15:00 in Area 2). Discharge is blocked below SoC 0.40, and a fixed reserve share $\rho = 0.3$ is held for frequency droop.

Day-ahead optimization (DAO): with perfect foresight of load, PV, and availability, the 24 hourly set-points of each area are optimized by Powell's method with the AC power flow and the fleet energy model inside the objective, minimizing the voltage, throughput, unmet-energy, and terminal-SoC terms of (21). The reserve share is fixed at $\rho = 0.3$. The search starts from an open-loop volt-watt schedule with fixed nighttime (and, in Area 2, PV-hour) charging and is limited to 20 Powell iterations; on the days checked, raising the limit to 80 iterations did not change the solution, but the method is local and the schedule is not guaranteed to be globally optimal. DAO serves as an informed benchmark for the voltage and energy part of the problem; it does not optimize the reserve allocation.

E. EVALUATION PROTOCOL AND METRICS

The test set contains 30 days drawn from the training distribution with an independent random stream, and the stress set contains 20 out-of-distribution days as defined above. For each controller and day the following daily metrics are computed: hours with any bus outside 0.95 to 1.05 p.u. (summed over the two areas); integrated undervoltage and overvoltage in p.u.·h summed over all buses; the daily minimum and maximum voltages; the mean over hours of the larger area frequency nadir; the worst nadir; the RMS frequency deviation; the maximum rate of change of frequency (RoCoF) over a 500-ms window; the RMS tie-line deviation; the area-level fleet throughput; the SoC deficit at the end of the day; unmet mobility energy; the battery-side regulation energy (7) under each treatment, including the exact 3600-s simulation; the numbers of raw-action clippings per action dimension and of projection events; the largest excess of the scheduled plus droop power over the charger and energy limits; the smallest sustained-activation energy margin; any SoC-limit violation; and the feeder energy loss. PPO results are pooled over the five seeds and the days, and seed-to-seed variability is reported separately as the standard deviation of the per-seed means.

IV. RESULTS AND DISCUSSION

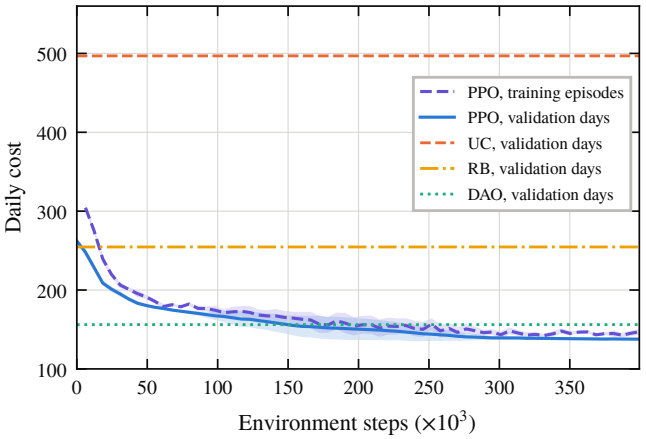


FIGURE 4. PPO learning curves (mean and ±1 standard deviation over 5 seeds) on training episodes and on ten fixed validation days, with the benchmark costs on the validation days.

A. MODEL VERIFICATION

The backward/forward sweep reproduces the published base case of the IEEE 33-bus feeder: the minimum voltage is 0.9131 p.u. at bus 18 and the active-power loss is 202.68 kW, in agreement with [2]. Over 20 random operating points with load factors between 0.4 and 1.1 and five random injections of up to ±0.8 MW, the largest voltage difference from the Newton–Raphson solution of pandapower [28] is 1.2×10^{-10} p.u. For the LFC model with AGC disabled, a 0.02-p.u. step in Area 1 gives a steady-state frequency deviation of -0.02381 Hz and a tie-line change of -5.000 MW, equal to the closed-form values $-\Delta P_L f_0/(B_1 + B_2) = -0.02381$ Hz and $-\Delta P_L B_2 P_r/(B_1 + B_2) = -5.000$ MW. With AGC active, both quantities return to zero, and the inter-area mode lies near 0.38 Hz. The verification scripts are part of the released code.

B. TRAINING BEHAVIOR

Fig. 4 shows the daily cost on the training episodes and on the ten validation days for the 5 seeds. Every seed is below the validation cost of the rule-based controller at the second evaluation (6144 steps) and falls below that of the day-ahead optimization between 104 448 and 264 192 steps. Over the final 50 000 steps, the validation cost of every seed changes by at most 1.0%, which satisfies the convergence criterion of Section III. The final validation cost is 137.7 ± 1.6 across seeds, compared with 156.1 for DAO, 254.6 for RB, and 496.9 for UC on the same days. The narrow seed-to-seed spread indicates that the result does not depend on a favorable initialization. Training one seed takes about 31 min on a single CPU core.

C. PERFORMANCE ON HELD-OUT TEST DAYS

Table 2 summarizes the 30 test days. PPO attains a mean daily cost of 135.5, which is 11.0% lower than DAO, 44.8% lower than RB, and 71.6% lower than UC. The seed-averaged PPO cost is lower than that of DAO on 30 of the 30 days and lower

than that of RB and UC on 30 and 30 days, respectively, and the paired Wilcoxon signed-rank test on the seed-averaged daily costs gives $p = 1.9 \times 10^{-9}$ against each benchmark.

Voltage. Uncontrolled charging coincides with the residential evening peak and drives the Area 1 voltage down to 0.835 p.u. on average, with 7.40 p.u.·h of integrated undervoltage per day. PPO reduces the undervoltage to 0.90 p.u.·h, which is 87.8% below UC, 68.4% below RB, and 13.5% below DAO, and it lowers the number of violation hours from 18.2 with DAO to 11.9. Most of the remaining violation hours are shallow (a few thousandths of a per unit) on the Area 1 feeder. Deeper violations remain in Area 2 between 18:00 and 21:00, when only a small part of the workplace fleet is still plugged in (Fig. 6). Overvoltage is rare for all controllers on the test days because the Area 2 workplace fleet absorbs midday PV; the PPO maximum voltage is 1.024 p.u.

Frequency. PPO holds an average of 25.3 MW of V2G reserve across both areas, against 15.2 MW for RB and 13.9 MW for DAO, which hold a fixed 30% share. The mean hourly nadir falls from 54.4 mHz with UC to 47.3 mHz, and it is 5.0% lower than with DAO. The maximum RoCoF decreases by 14.1% relative to UC. These gains are modest in absolute terms because the V2G reserve is a small fraction of the 500-MW area base and the governors respond within about one second. They are nonetheless obtained without degrading the voltage objective, which is the point of the joint allocation.

Energy and fleet. PPO leaves a terminal SoC deficit of 6.31%, against 9.68% for DAO and 13.58% for RB. Its fleet throughput, 446 MWh at area level, is higher than that of DAO (348 MWh) and RB because the policy charges during the day to supply the evening discharge. Section IV-D examines the energy and power limits in detail.

D. REGULATION ENERGY, TREATMENT SENSITIVITY, AND FEASIBILITY AUDIT

Physical interpretation of the hourly extrapolation. The 60-s window of each hour contains one step load change at $t = 2$ s and a continuous fluctuation. Multiplying the window energy by 60 therefore assumes that the step event recurs every minute, whereas the model defines one step per hour. The exact 3600-s simulations recorded for every evaluated hour show how large the error is. For the PPO policies the gross regulation energy over the test days is 8.84 MWh per day in the exact simulation, against 25.44 MWh with the hourly extrapolation (2.9 times too large) and 0.42 MWh with the window alone (more than 20 times too small). The decomposed treatment (8) gives 8.29 MWh, or 0.94 of the exact value. The same ratio of 0.94 to 0.94 is found for RB and DAO, so the remaining 6% shortfall is systematic rather than random. A likely cause, which we state as an interpretation, is that the fluctuation-only window starts from rest, so that its first seconds under-represent the stationary fluctuation before the scaling to 59 minutes. The net daily energies are -0.73 MWh (exact), -0.69 MWh (decomposed), -1.67 MWh (hourly extrapolation), and -0.03 MWh (window). The hourly extrapolation also gets the sign of the hourly net energy wrong in 39% of the hours, because the sign of the single step event dominates the window while the full hour is dominated by the fluctuation and by the lossy round trips. The same ratios are found for RB and DAO (gross ratio 2.7 and 2.6 for the extrapolation, 0.94 and 0.94 for the decomposition). Repeating the 60-s response for an entire hour is therefore not a physically justified treatment: it overstates both the gross and the net regulation energy by a factor of about two to three, and it is not simply conservative, because its error has either sign hour by hour. The decomposed treatment is closer to the exact simulation than any other cheap treatment but underestimates the gross energy by about 6%.

Sensitivity of the results to the treatment. Table 3 evaluates every controller on the same 30 test days under the five treatments. For PPO, the daily cost changes by at most 0.1 between any treatment and the default, which is 0.1 times the seed-to-seed standard deviation of the final policies (1.3); the undervoltage and the violation hours change by at most 0.001 p.u.·h and 0.1 h, the nadir is unchanged to the reported precision, and the terminal SoC deficit moves between 6.26% (neglected) and 6.42% (hourly extrapolation). No treatment produces an SoC-limit violation or any additional unmet mobility energy for any controller; the unmet energy of DAO (1.28 MWh) is caused by its schedule and is the same under all treatments to within 0.03 MWh. The regulation-energy assumption therefore does not change the control problem, the ranking of the controllers, or any conclusion of this paper. The decomposed treatment is retained as the default because it tracks the exact simulation to within 6% at the cost of one additional 60-s simulation per hour.

Magnitude of the term. Table 4 reports the regulation energy of the default treatment on the test days. For PPO it averages 0.091 MWh per hour in absolute value over both areas, with a maximum of 0.607 MWh in one hour; the gross daily energy is 0.51% of the battery energy moved by the scheduled set-points, and the largest net change of one area fleet over a day is 0.38 percentage points of its capacity. The mean net energy is negative for every controller that holds reserve, because each round trip is lossy, but 13% of the PPO seed-days end with a small net gain, depending on the signs of the load steps.

Feasibility under sustained activation. The audit columns of Table 4 and the exact simulations confirm the guarantees of Section III-A. Over all test and stress days and all treatments, the scheduled power plus the instantaneous droop power never exceeds the charger or energy limits (largest excess 0.000 MW), the droop power never exceeds the committed reserve, no SoC limit is violated, and the energy balance (6) closes to within 3.6e-15 MWh in every hour. The sustained-activation margin, that is, the energy that would remain above $E^{\min}$ (or below $E^{\max}$) if the full reserve were activated in one direction for the whole hour, is never negative. Its minimum is exactly zero: for PPO, (19) or (20) limits the reserve in 0.1 feeder-hours per day on average, and for DAO the zero margins (2.8 feeder-hours per day) all occur in hours with

TABLE 2. Results on the 30 held-out test days (mean ± standard deviation across days). PPO values are pooled over 5 seeds; the value in parentheses is the standard deviation of the per-seed means.

Metric	UC	RB	DAO	PPO (proposed)
Daily cost $\sum_t c_t$	477.2 ± 74.8	245.5 ± 31.6	152.3 ± 31.8	135.5 ± 26.0 (1.3)
Voltage-violation hours (h)	19.3 ± 3.4	12.5 ± 2.1	18.2 ± 4.0	11.9 ± 5.5 (1.1)
Undervoltage (p.u.·h)	7.40 ± 1.49	2.86 ± 0.59	1.04 ± 0.37	0.90 ± 0.35 (0.01)
Overvoltage (p.u.·h)	0.050 ± 0.082	0.001 ± 0.005	0.001 ± 0.005	0.004 ± 0.016 (0.001)
Minimum voltage (p.u.)	0.835 ± 0.015	0.907 ± 0.007	0.922 ± 0.007	0.925 ± 0.006 (0.000)
Maximum voltage (p.u.)	1.044 ± 0.028	1.024 ± 0.016	1.029 ± 0.019	1.024 ± 0.022 (0.000)
Mean hourly nadir (mHz)	54.4 ± 6.2	49.4 ± 6.0	49.8 ± 6.0	47.3 ± 5.9 (0.1)
Worst nadir (mHz)	79.2 ± 10.4	70.5 ± 7.6	70.5 ± 7.6	69.6 ± 7.5 (0.4)
RMS frequency deviation (mHz)	12.80 ± 1.22	11.48 ± 1.16	11.57 ± 1.18	10.87 ± 1.10 (0.01)
Maximum RoCoF (Hz/s)	0.154 ± 0.024	0.137 ± 0.016	0.137 ± 0.016	0.133 ± 0.015 (0.001)
RMS tie-line deviation (MW)	2.51 ± 0.23	2.49 ± 0.24	2.49 ± 0.24	2.51 ± 0.25 (0.00)
Mean V2G reserve, both areas (MW)	0.0 ± 0.0	15.2 ± 1.4	13.9 ± 2.6	25.3 ± 2.8 (0.3)
Fleet throughput, area level (MWh)	491 ± 19	337 ± 92	348 ± 26	446 ± 39 (3)
Terminal SoC deficit (%)	0.00 ± 0.00	13.58 ± 14.06	9.68 ± 13.55	6.31 ± 6.80 (0.69)
Projection events per day	27.8 ± 5.0	0.0 ± 0.0	1.6 ± 2.8	0.0 ± 0.0 (0.0)
Energy loss, two feeders (MWh)	4.978 ± 0.595	3.416 ± 0.261	3.217 ± 0.338	3.159 ± 0.357 (0.015)
Unmet mobility energy (MWh)	0.00 ± 0.00	0.00 ± 0.00	1.28 ± 2.55	0.00 ± 0.00 (0.00)

TABLE 3. Sensitivity of the daily results on the 30 test days to the treatment of the regulation energy. Every row uses the same scenarios, disturbances and controllers; only the energy credited to the droop loop in (6) differs. PPO rows pool the five final policies. The last column is the energy by which the stored energy would have left [$E^{\min}$, $E^{\max}$] without the limit.

Controller	Treatment	Cost	SoC def. (%)	Unmet (MWh)	UV (p.u.·h)	Viol. h	Nadir (mHz)	Reg. energy (MWh/d)	Limit viol. (MWh)
UC	None (neglected)	477.2	0.00	0.00	7.402	19.3	54.4	0.00	0.000
	60-s window only	477.2	0.00	0.00	7.402	19.3	54.4	0.00	0.000
	Hourly extrapolation (×60)	477.2	0.00	0.00	7.402	19.3	54.4	0.00	0.000
	Decomposed (default)	477.2	0.00	0.00	7.402	19.3	54.4	0.00	0.000
	Exact 3600-s simulation	477.2	0.00	0.00	7.402	19.3	54.4	0.00	0.000
RB	None (neglected)	245.0	13.71	0.00	2.848	12.5	49.4	0.00	0.000
	60-s window only	245.0	13.72	0.00	2.848	12.5	49.4	-0.02	0.000
	Hourly extrapolation (×60)	247.0	13.66	0.00	2.887	12.6	49.4	-1.10	0.000
	Decomposed (default)	245.5	13.58	0.00	2.860	12.5	49.4	-0.46	0.000
	Exact 3600-s simulation	245.1	13.75	0.00	2.848	12.5	49.4	-0.50	0.000
DAO	None (neglected)	152.2	9.59	1.28	1.045	18.2	49.8	0.00	0.000
	60-s window only	152.2	9.59	1.28	1.045	18.2	49.8	-0.02	0.000
	Hourly extrapolation (×60)	152.5	9.92	1.30	1.045	18.2	49.8	-1.02	0.000
	Decomposed (default)	152.3	9.68	1.28	1.045	18.2	49.8	-0.43	0.000
	Exact 3600-s simulation	152.3	9.68	1.28	1.045	18.2	49.8	-0.47	0.000
PPO (proposed)	None (neglected)	135.5	6.26	0.00	0.904	11.9	47.3	0.00	0.000
	60-s window only	135.5	6.26	0.00	0.904	11.9	47.3	-0.03	0.000
	Hourly extrapolation (×60)	135.6	6.42	0.00	0.905	12.0	47.3	-1.67	0.000
	Decomposed (default)	135.5	6.31	0.00	0.904	11.9	47.3	-0.69	0.000
	Exact 3600-s simulation	135.5	6.30	0.00	0.904	11.9	47.3	-0.73	0.000

TABLE 4. Energy exchanged by the V2G droop loop and feasibility audit on the test days (area level, both areas). Regulation energy is the battery-side energy including η_{ch} and η_{dis}; negative values reduce stored energy. Values use the default decomposed treatment (8); the window-only column counts the simulated 60 s. The share is the gross (absolute) regulation energy relative to the battery energy of the scheduled set-points. Counts are summed over all test days (and seeds for PPO); action-clipping counts refer to individual action components.

	Regulation energy					Feasibility audit			
Controller	Mean | · | per h (MWh)	Max | · | per h (MWh)	Daily sum (MWh)	Daily sum, window only (MWh)	Gross share of sched. (%)	Raw action clippings	Power projections	Max power excess (MW)	Max SoC-limit violation (MWh)
UC	0.000	0.000	0.00 ± 0.00	0.0000	0.00	0	833	0.000	0.000
RB	0.059	0.398	-0.46 ± 0.44	-0.0184	0.43	0	0	0.000	0.000
DAO	0.056	0.398	-0.43 ± 0.43	-0.0172	0.40	0	48	0.000	0.000
PPO (proposed)	0.091	0.607	-0.69 ± 0.68	-0.0278	0.51	7080	0	0.000	0.000

a mobility shortfall, where no upward reserve is credited by design. In the exact 3600-s simulations the droop power reaches its saturation limit in 0.00% of the simulated time, so the actual activation stays far below the sustained worst case that the design limit protects against. Projection events occur for UC (833 on the test days), which requests full charging until the fleet is full, and occasionally for DAO (48), whose open-loop schedule does not see the realized regulation energy; they do not occur for RB or PPO.

E. RAW-ACTION SATURATION AND A BOUNDED-ACTION POLICY

Table 5 reports, for each action dimension, how often the deterministic action of the final policies lies outside $[-1, 1]$ before clipping. Over the 3 600 hourly decisions of the five policies on the test days, 49.2% of the action components are saturated: 38.5% for the Area 1 set-point (13.4% above, 25.1% below), 55.5% for the Area 2 set-point, 54.0% for the Area 1 reserve share (mostly above, that is, at $\rho_{\max}$) and 48.6% for the Area 2 share. The median excess beyond the bound is 0.61. These saturations correspond to charging or discharging at the full available rating, or to the minimum or maximum reserve share, which are admissible operating points because the action bounds coincide with the design limits; none of them is a physically infeasible request, as the zero projection count for PPO shows. The high rate does, however, mean that the Gaussian mean often lies well outside the box, so the gradient of the clipped action with respect to the policy parameters is zero in those states and exploration there is wasted.

To test whether a properly bounded policy performs better, a policy with a hyperbolic-tangent squashed output and generalized state-dependent exploration [33] was trained with three seeds and the same 200 000-step budget as the ablation reference, with all other hyperparameters unchanged. Its action is bounded by construction (saturation 0.0%). Its test cost is 178.4 ± 0.4 against 147.3 ± 10.7 for the clipped Gaussian reference (seed costs 178.7, 178.4, 178.0 versus 140.5, 159.7, 141.8), with higher undervoltage (1.18 p.u.·h against 0.92) and a larger terminal SoC deficit (28.14% against 15.62%). Its learning curves plateau early, at a validation cost near the final value from about 50 000 steps, and change by at most 0.4% over the last 50 000 steps, whereas the clipped reference is still improving at the same point. Our interpretation, which we have not tested further, is that the squashing compresses the policy gradient near the bounds, which are precisely the operating points that the task rewards, so that the bounded policy settles in a worse region of the decision space. We therefore keep the clipped Gaussian policy for the main results, report its saturation statistics openly, and note that a longer training budget, different exploration settings, or a Beta-distributed policy could change this comparison.

F. OPERATING BEHAVIOR ON A REPRESENTATIVE DAY

Fig. 5 shows the first test day. Under UC, the Area 1 minimum voltage falls to 0.84 p.u. at 19:00 because arriving vehicles charge at full rate during the peak. RB discharges once the forecast voltage crosses its threshold, but its night-time charging at full rate pulls the voltage below 0.91 p.u. between midnight and 05:00. DAO keeps the voltage close to the 0.95 p.u. limit throughout, as its foresight allows. PPO begins discharging in Area 1 at about 17:00, when the minimum voltage reaches the 0.95 p.u. limit, reaches about 19 MW at 19:00, and recharges at a moderate rate overnight, so the minimum voltage stays near 0.95 p.u. and the Area 1 SoC ends the day close to its initial value. In Area 2, PPO charges while the workplace fleet is plugged in (06:00 to 15:00, largely from PV), starts discharging at 16:00, and discharges at the full plugged-in rating at 17:00, just before the fleet leaves. Fig. 5(f) shows that PPO keeps 23 to 33 MW of reserve for most of the day and reduces it at 17:00 to 19:00, when the converter capacity is needed for discharge. The voltage maps in Fig. 6 show that the deep undervoltage on the main feeder and the bus 26 to 33 lateral of Area 1 is removed, while the evening undervoltage in Area 2 is reduced but not eliminated.

G. FREQUENCY RESPONSE WITHIN AN HOUR

Fig. 7 shows the 60-s window at 13:00 of the same day. PV has reduced the inertia constant of Area 2 to 3.25 s, and a load step occurs in Area 1 at $t = 2$ s; Area 2 sees the disturbance through the tie line. Without reserve (UC), the Area 2 nadir reaches 76.9 mHz and the inter-area oscillation is lightly damped. PPO holds 32.3 MW of reserve at this hour, most of it in Area 2 where the workplace fleet is connected, against 19.1 MW for RB. The Area 2 nadir is reduced to 47.7 mHz (57.0 mHz with RB) and the oscillation decays faster. The Area 1 nadir changes much less (78.6, 73.5, and 74.4 mHz for UC, RB, and PPO), because only about a quarter of the residential fleet is plugged in at midday and PPO assigns it a small reserve share. The worst nadir over the two areas in this hour is therefore 74.4 mHz for PPO and 73.5 mHz for RB. This example shows that the benefit of reserve depends on where the disturbance occurs relative to where the reserve is held, which is why the test-set gains in Table 2 are averages over random disturbance locations.

H. LEARNED ALLOCATION BETWEEN VOLTAGE SUPPORT AND RESERVE

Fig. 8 averages the PPO decisions over all test days and seeds. The reserve share follows fleet availability more than inertia. In Area 2, the policy holds $\rho \approx 0.58$ between 10:00 and 14:00, when the workplace fleet is connected and PV has lowered the inertia, and releases the reserve to zero in the evening, when the few remaining vehicles are discharged to support the feeder voltage. In Area 1, the reserve share is close to its upper limit at night ($\rho \approx 0.60$), when inertia is high but the residential fleet is plugged in and otherwise idle, drops to 0.11 at midday, when most residential vehicles are away and the remaining ones charge, and settles near 0.46 during the evening peak, when part of the converter capacity is used for discharge. The set-points show charging from midday PV in both areas, discharge in the evening peak, and moderate overnight charging in Area 1. None of these rules was specified in the reward; they emerge from the trade-off encoded in (21).

Fig. 9 shows the distributions behind the averages. The undervoltage distribution of PPO lies below those of RB and DAO, and its throughput is higher. Its mean SoC deficit is lower than that of RB and DAO, but the median is not: DAO ends 3% of the days with no deficit (median 0.21%), whereas PPO tends to leave a small deficit on many days (median 4.23%). The nadir distributions of all controllers overlap,

TABLE 5. Raw-action saturation per action dimension on the test days (percentage of hourly decisions whose deterministic action lies outside $[-1, 1]$ before clipping; for the squashed policy the action is bounded by construction) and the resulting performance. Seeds in parentheses.

Policy (seeds)	a_1 (P_1)	a_2 (P_2)	a_3 (ρ_1)	a_4 (ρ_2)	All	Cost	UV (p.u.·h)	Nadir (mHz)	Reserve (MW)	SoC def. (%)
Clipped Gaussian (reference, 200k) (3)	38.3	50.5	41.5	37.6	42.0	147.3 ± 10.7	0.92	47.8	22.9	15.62
Squashed Gaussian + gSDE (200k) (3)	0.0	0.0	0.0	0.0	0.0	178.4 ± 0.4	1.18	48.9	18.6	28.14
Clipped Gaussian (final, 400k) (5)	38.5	55.5	54.0	48.6	49.2	135.5 ± 1.3	0.90	47.3	25.3	6.31

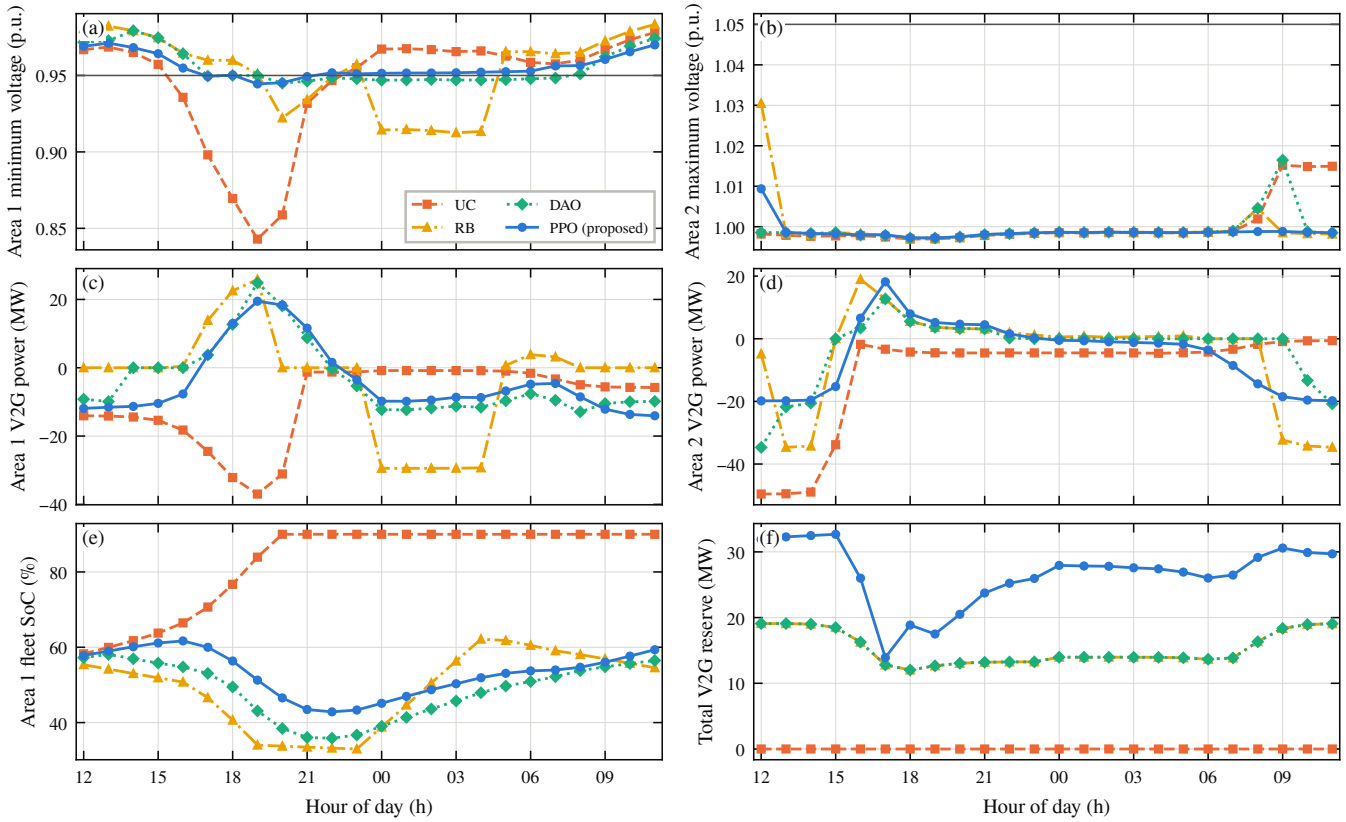


FIGURE 5. Representative test day (day 1, PPO seed 0). (a) Area 1 minimum bus voltage. (b) Area 2 maximum bus voltage. (c), (d) Area-level V2G set-points (positive values denote discharge). (e) Area 1 fleet SoC. (f) Total V2G frequency reserve.

because the day-to-day variation of the disturbance size is larger than the effect of the reserve; compared day by day, however, the PPO mean nadir is lower than that of DAO on 30 of the 30 test days, than that of RB on 30, and than that of UC on 30 (paired Wilcoxon test against DAO, $p = 1.9 \times 10^{-9}$).

I. OUT-OF-DISTRIBUTION STRESS DAYS

Table 6 reports the 20 stress days, which combine 10% to 20% heavier load, 30% to 45% lower fleet participation, and larger disturbances, all beyond the training ranges, with clear-sky PV. All controllers deteriorate. PPO remains the lowest-cost controller, 8.0% below DAO and 23.2% below RB, and its daily cost is the lowest on all 20 days ($p = 1.9 \times 10^{-6}$ against each benchmark). Its undervoltage is 7.7% below DAO. The policy does, however, show several signs of operating outside its training range. It records more violation hours than RB (18.5 against 14.2), because it spreads the limited fleet energy over more hours with shallow violations, its terminal SoC deficit rises to 18.02%, and its maximum voltage reaches 1.052 p.u. Training on a wider scenario distribution, or adding a hard terminal-energy constraint, would be needed before deployment under such conditions.

J. REWARD ABLATION AND WEIGHT SENSITIVITY

Tables 7 and 8 report policies trained with one reward term removed, or with the voltage or frequency weight halved and doubled, and evaluated with the nominal cost (21). To limit computation each variant was trained for 200 000 steps, with three seeds for the ablation and two for the sensitivity study, and a reference policy with the nominal reward was trained under the same budget in the same environment. The comparison is made against this reference, not against the 400 000-step policies of Table 2. The same seed means the same initial network weights and the same scenario stream, so the seed-paired difference between a variant and the reference isolates the effect of the reward change.

Frequency terms. Removing them has a clear and consistent effect. All three seeds give up most of the reserve

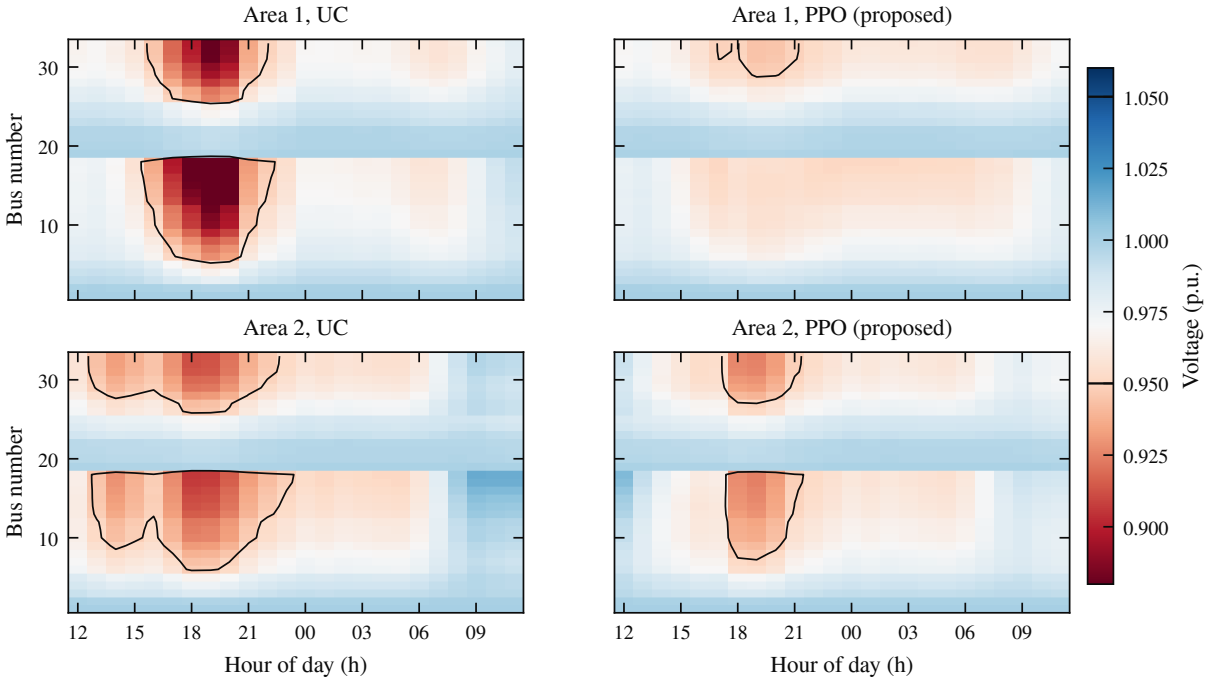


FIGURE 6. **Bus voltages over the representative test day under UC and PPO. Black contours mark 0.95 and 1.05 p.u.**

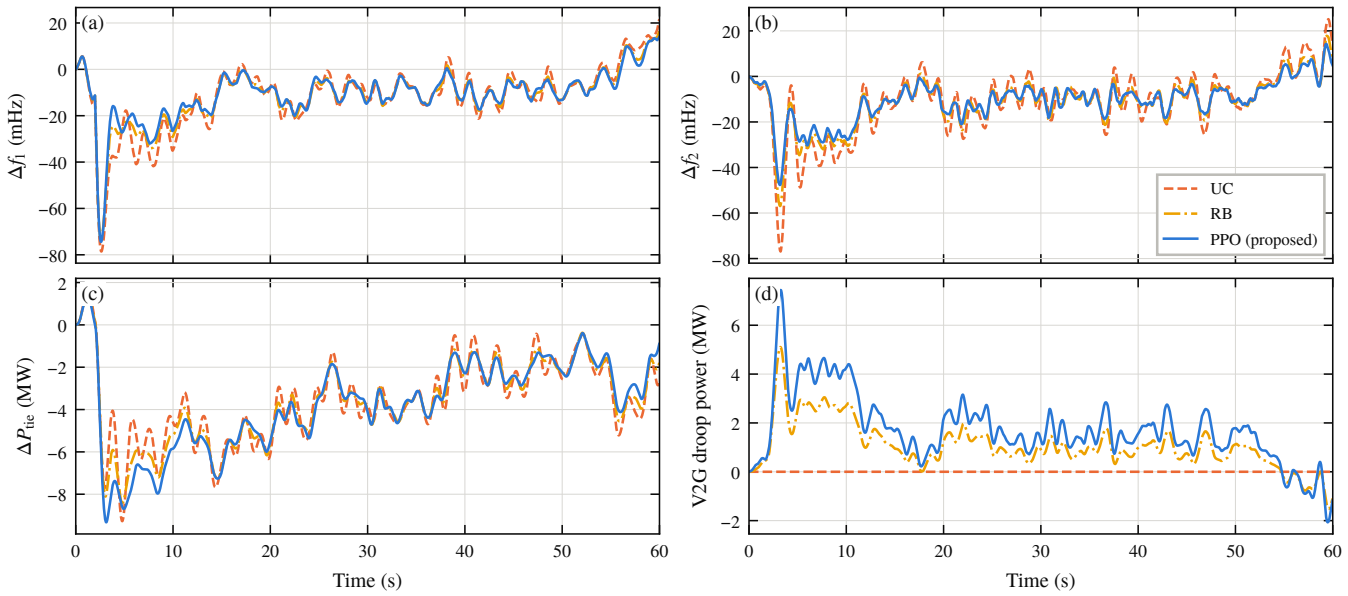


FIGURE 7. **Frequency response at 13:00 of the representative day. (a), (b) Area frequency deviations. (c) Tie-line power deviation. (d) Total V2G droop power.**

(10.1 MW instead of 22.9 MW), raise the mean nadir from 47.8 to 51.0 mHz, and increase the nominal cost in every seed pair (differences +29.4, +8.4, +27.3; mean 21.7). Halving w_f moves the policy in the same direction (reserve 16.5 MW, paired cost differences +17.3, +6.3), and doubling it raises the reserve to 26.6 MW at the price of more undervoltage (1.00 p.u.·h against 0.92). The reserve allocation of Fig. 8 is therefore controlled by the frequency weight, and the nominal $w_f = 4$ sits between the two extremes.

Voltage weight. Doubling w_V reduces undervoltage to 0.85 p.u.·h but cuts the reserve by 39% (13.9 MW) and raises the terminal SoC deficit to 30.85%, because the policy discharges more in the evening and recharges less. Halving w_V does the opposite. The two weights thus trade voltage support against reserve and end-of-day energy in the expected directions.

Why some variants show a lower total cost than the reference. Removing the overvoltage or throughput term, halving

TABLE 6. Results on the 20 out-of-distribution stress days (mean ± standard deviation across days). PPO values are pooled over 5 seeds; the value in parentheses is the standard deviation of the per-seed means.

Metric	UC	RB	DAO	PPO (proposed)
Daily cost $\sum_t c_t$	671.4 ± 24.3	369.4 ± 17.6	308.3 ± 19.6	283.8 ± 16.2 (3.1)
Voltage-violation hours (h)	24.3 ± 2.3	14.2 ± 1.3	18.6 ± 4.5	18.5 ± 3.0 (1.0)
Undervoltage (p.u.·h)	10.25 ± 0.50	4.15 ± 0.36	2.88 ± 0.29	2.66 ± 0.27 (0.13)
Overvoltage (p.u.·h)	0.070 ± 0.024	0.000 ± 0.001	0.000 ± 0.001	0.006 ± 0.009 (0.003)
Minimum voltage (p.u.)	0.835 ± 0.006	0.898 ± 0.003	0.903 ± 0.004	0.905 ± 0.004 (0.000)
Maximum voltage (p.u.)	1.070 ± 0.004	1.043 ± 0.005	1.048 ± 0.002	1.052 ± 0.005 (0.002)
Mean hourly nadir (mHz)	82.6 ± 6.4	76.8 ± 6.2	77.6 ± 6.3	75.0 ± 6.1 (0.4)
Worst nadir (mHz)	121.3 ± 11.5	108.6 ± 9.9	108.6 ± 10.0	105.8 ± 9.3 (0.5)
Maximum RoCoF (Hz/s)	0.240 ± 0.021	0.215 ± 0.018	0.215 ± 0.018	0.205 ± 0.019 (0.002)
Terminal SoC deficit (%)	0.00 ± 0.00	26.16 ± 3.65	24.30 ± 8.70	18.02 ± 4.07 (2.94)
Projection events per day	16.4 ± 1.0	0.0 ± 0.0	2.8 ± 3.0	0.0 ± 0.0 (0.0)
Unmet mobility energy (MWh)	0.00 ± 0.00	0.00 ± 0.00	2.43 ± 2.76	0.00 ± 0.00 (0.00)

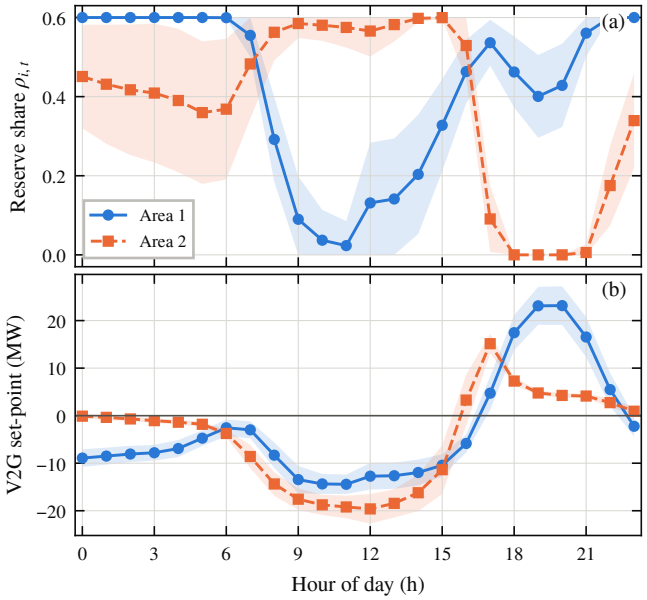


FIGURE 8. PPO decisions averaged over the test days and 5 seeds (mean and ±1 standard deviation). (a) Reserve share. (b) Area-level V2G set-point.

w_V, or doubling w_f yields a lower mean nominal cost than the reference (Tables 7 and 8), which at first sight suggests that these terms are harmful. The cost decomposition shows otherwise. The voltage, frequency, and throughput components of the reference and of these variants differ by only a few units (voltage 46.4 against 46.4 and 46.3; frequency 80.3 against 79.4 and 80.2). Almost the entire difference comes from the terminal SoC term, which is 15.6 for the reference, 13.8 without the overvoltage term, 13.9 without the throughput term, 2.2 with $w_V = 25$, and 7.8 with $w_f = 8$. This term is paid once per episode and is the slowest part of the reward to learn: at 200 000 steps the three reference seeds leave terminal deficits of 7.7, 27.8, 11.4 percentage points (seed costs 140.5, 159.7, 141.8), whereas the 400 000-step policies of Table 2 reach 6.31%. The apparent advantage of the variants is therefore a training-budget effect on a term unrelated to the removed one, and it is not statistically distinguishable from the reference (Welch tests on seed means, $p = 0.70$ and $p = 0.85$; paired differences +0.5, −9.1, −0.3 and −1.0, −2.2, −2.0). Consistent with this, the 200 000-step runs with large terminal deficits have not met the convergence criterion of Section III-C (validation cost still changing by up to 11.4% over the final 50 000 steps). The overvoltage term has little effect on the test days because overvoltage is rare for every policy, and the throughput term with $w_{\mathrm{deg}} = 0.05$ is a minor part of the cost. We keep both terms because they encode constraints that matter on days the test set does not represent, but the evidence here does not show that they improve performance.

K. LIMITATIONS

Several simplifications bound the conclusions. Each area's distribution system is represented by identical copies of one feeder, so feeder diversity and the coupling between distribution voltages and transmission-level frequency are not captured. The fleet is modeled as one aggregated battery per feeder; individual departure constraints, charger heterogeneity, and battery aging physics are not modeled, and the throughput term is only a proxy for degradation. Reactive-power capability of the chargers, which is effective for voltage control [11], is not used. The load, PV, and availability profiles are synthetic, and the LFC model is linearized with representative parameters rather than identified from a specific system. Finally, the agent receives hour-ahead forecasts without error; forecast uncertainty and communication delays at the aggregator level are left for future work.

V. CONCLUSION

This paper formulated the coordination of V2G voltage support and frequency reserve as a two-time-scale control problem and solved it with PPO. The hourly decision of each area aggregator consists of a fleet set-point and a reserve share. The set-point is evaluated by AC power flow on IEEE 33-bus feeders and an energy-consistent fleet model, and the reserve determines the gain and saturation of a V2G droop loop in a dynamic two-area LFC model with inertia, governors, a tie line, and AGC. All quantities are expressed in physical units,

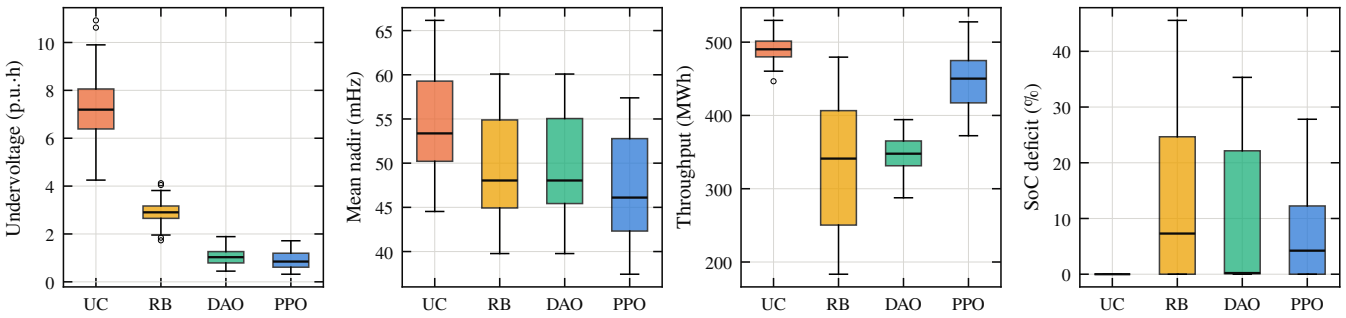


FIGURE 9. **Distribution of daily metrics over the test days (PPO pooled over 5 seeds).**

TABLE 7. **Reward ablation on the test days (mean ± standard deviation across seeds). Every policy is trained for 200 000 steps in the corrected environment with the modified reward and evaluated with the full cost** (21)**.**

Reward variant (seeds)	Cost	UV (p.u.·h)	OV (p.u.·h)	Nadir (mHz)	Reserve (MW)	Thr. (MWh)	SoC def. (%)
Full reward (reference) (3)	147.3 ± 10.7	0.92 ± 0.03	0.003 ± 0.002	47.8 ± 0.4	22.9 ± 2.6	437 ± 25	15.62 ± 10.68
$w_{\mathrm{ov}} = 0$ (no overvoltage term) (3)	144.4 ± 5.4	0.92 ± 0.02	0.004 ± 0.001	47.5 ± 0.2	24.4 ± 0.5	431 ± 32	13.75 ± 7.06
$w_f = w_{\mathrm{tie}} = 0$ (no frequency terms) (3)	169.0 ± 0.9	0.86 ± 0.01	0.001 ± 0.000	51.0 ± 0.4	10.1 ± 1.3	512 ± 29	28.68 ± 0.36
$w_{\mathrm{deg}} = 0$ (no throughput term) (3)	145.6 ± 10.3	0.92 ± 0.03	0.002 ± 0.001	47.8 ± 0.3	23.1 ± 1.9	452 ± 23	13.88 ± 10.75

TABLE 8. **Reward-weight sensitivity on the test days (mean ± standard deviation across seeds; 200 000 steps; evaluation with the nominal weights of Table 1).**

Reward variant (seeds)	Cost	UV (p.u.·h)	OV (p.u.·h)	Nadir (mHz)	Reserve (MW)	Thr. (MWh)	SoC def. (%)
Full reward (reference) (3)	147.3 ± 10.7	0.92 ± 0.03	0.003 ± 0.002	47.8 ± 0.4	22.9 ± 2.6	437 ± 25	15.62 ± 10.68
$w_V = 25$ (half) (2)	140.1 ± 0.4	1.09 ± 0.03	0.008 ± 0.003	47.0 ± 0.1	26.4 ± 0.4	428 ± 3	2.17 ± 0.78
$w_V = 100$ (double) (2)	167.5 ± 0.4	0.85 ± 0.01	0.003 ± 0.002	49.9 ± 0.1	13.9 ± 0.3	442 ± 19	30.85 ± 0.28
$w_f = 2$ (half) (2)	161.9 ± 5.8	0.87 ± 0.00	0.001 ± 0.001	49.3 ± 0.8	16.5 ± 3.4	458 ± 30	27.38 ± 2.56
$w_f = 8$ (double) (2)	140.5 ± 0.4	1.00 ± 0.01	0.007 ± 0.001	46.9 ± 0.4	26.6 ± 1.3	415 ± 21	7.84 ± 0.90

and the models were verified against pandapower and closed-form results.

Across 5 training seeds and 30 held-out days, the learned policy reduced the daily cost by 11.0% relative to a perfect-foresight day-ahead optimization with fixed reserve, by 44.8% relative to a rule-based volt-watt and droop controller, and by 71.6% relative to uncontrolled charging. Undervoltage fell by 13.5% relative to the day-ahead optimization, the mean frequency nadir by 5.0%, and the terminal SoC deficit from 9.68% to 6.31%. The policy learned to hold reserve mainly where and when vehicles are plugged in, including the low-inertia midday hours in the workplace area, and to release it when converter capacity is needed for voltage support. On out-of-distribution stress days the policy kept the lowest cost but showed larger SoC deficits and more frequent shallow violations, which marks the limits of the training distribution.

Future work will add charger reactive power, forecast uncertainty, heterogeneous feeders, electricity-cost terms in the objective, and constrained RL formulations that enforce terminal energy requirements as hard constraints.

DATA AND CODE AVAILABILITY

The simulation environment, training and evaluation scripts, trained policies, and the data behind every table and figure are available at [repository URL and DOI to be added].

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The authors used an AI assistant (Claude, Anthropic) to help implement and debug the simulation code and to edit the language of the manuscript. All models, parameters, results, and interpretations were checked by the authors, who take full responsibility for the content.

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